Quantifying Ethylene Oxide-Induced DNA Alkylation Risks: Spatial Modeling of Industrial Sterilization Emissions and Modeled Cancer Risk in New Jersey Communities

1. Executive Summary & Abstract

1.1 Abstract

Industrial ethylene oxide (EtO) emissions present significant public health concerns due to the chemical's ability to directly damage DNA through alkylation and increase lifetime cancer risk. Traditional municipal zoning and county-level averages often fail to capture micro-spatial variations and localized pollution hotspots. To address this problem, this study developed a spatial computational framework using QGIS (v3.x) to integrate 2020 U.S. Census block geometries from the EtO_Risk_NJ_2020 dataset with industrial sterilization facility locations from the EtO_Facilities_NJ dataset across all 21 New Jersey counties. The spatial and proximity analysis revealed extreme spatial heterogeneity, uncovering severe urban hotspots with modeled risks reaching up to 567.27 per million in Union County and 230.05 per million in Middlesex County, as well as a regional cross-border plume in southwestern New Jersey where Salem County showed an elevated mean risk of 5.18 per million. Quantitative statistical analysis of buffer zones demonstrated that a 3.11-mile (5 km) radius serves as an optimal screening boundary, capturing 98.4% of census blocks with risks exceeding the EPA benchmark of 1.0 per million. For future directions, this spatial screening tool can be integrated with multi-year Toxics Release Inventory (TRI) emission scaling and real-time fenceline monitoring to improve land-use planning, address environmental justice concerns, and guide targeted public health investigations.

1.2 Statement of Hypothesis

If atmospheric dispersion patterns and toxicity estimates are mapped around industrial ethylene oxide (EtO) sterilization facilities, then census blocks located closer to these facilities, including rural border areas such as Salem County that may be influenced by emissions from nearby out-of-state facilities, will show higher modeled lifetime cancer-risk values - serving as a population-level proxy for underlying DNA alkylation risk - than areas farther away. This would indicate that high-resolution spatial analysis can identify localized areas of elevated risk that may not be apparent from standard zoning or county-level averages.

2. Introduction & Literature Review

2.1 Molecular Biochemistry of Ethylene Oxide Toxicity

Ethylene oxide (EtO) is a highly reactive, ring-shaped chemical that is widely used in industrial manufacturing and to sterilize medical equipment. Its strained three-membered ring makes it especially reactive with biological molecules. When EtO is inhaled, it can enter the body and react

directly with DNA through a process called alkylation, meaning that EtO can damage DNA without first needing to be converted into another chemical by the body.

One of the major DNA changes caused by EtO is the formation of N7-(2-hydroxyethyl)guanine (N7-HEG), which accounts for approximately 75–80% of the DNA alkylation products produced by EtO exposure (Marsit et al., 2016). Other DNA adducts can also form, including N3-(2-hydroxyethyl)adenine. If these damaged DNA bases are not properly repaired through the cell's base excision repair (BER) system, they can interfere with normal DNA replication and repair. Over time, this damage may contribute to chromosomal abnormalities and mutations in genes that regulate cell growth, including important oncogenes and tumor suppressor genes such as *KRAS* and *TP53* (ATSDR, 2022; IARC, 2012; Marsit et al., 2016).

Mutations affecting genes such as *TP53*, which helps control DNA damage responses and prevent abnormal cell growth, and *KRAS*, which regulates cell-growth signaling, can contribute to the development and progression of cancer. Because of these molecular effects, long-term EtO exposure has been associated with an increased risk of lymphohematopoietic cancers, including leukemia and lymphoma, as well as breast cancer. Understanding how EtO causes DNA damage at the molecular level is therefore important for evaluating how exposure may translate into health risks for nearby communities.

2.2 The Spatial Problem and Environmental Justice

Current municipal zoning regulations often rely on broad geographic boundaries, such as county or township averages, when evaluating air pollution exposure. However, airborne chemicals do not spread evenly across these large areas. Their concentrations can change significantly over short distances depending on factors such as stack height, the speed and direction of emissions, local terrain, and small-scale wind patterns. Because of this, standard air monitoring networks may miss highly localized pollution hotspots, particularly areas within a few hundred meters of facility boundaries or communities located downwind of pollution sources across state lines.

Industrial sterilization and chemical manufacturing facilities can release a mixture of volatile organic compounds (VOCs) and other pollutants. For this spatial EtO cancer-risk model, however, ethylene oxide (EtO) was selected as the primary pollutant and toxicological focus because of its highly reactive chemical structure and its ability to directly damage DNA through alkylation. EtO can form N-(2-hydroxyethyl)guanine (N7-HEG) adducts without requiring metabolic activation, making its mechanism of DNA damage particularly relevant to this study. Based on EPA health-risk assessments, EtO is also considered a major contributor to excess cancer risk associated with certain industrial exposure scenarios. For these reasons, EtO was used as the primary pollutant and toxicological focus for identifying and evaluating localized community exposure patterns.

3. Materials & Methods

3.1 Data Architecture & Integration

To build a high-resolution spatial model covering all 21 counties in New Jersey, I combined two main data layers using the open-source GIS software QGIS (v3.x). The datasets were processed

and compiled into an offline GeoPackage database so that the analysis could be performed consistently across the state.

- **Point-Source Facility Layer (EtO_Facilities_NJ):** This layer was created from the EPA Toxics Release Inventory (TRI) dataset (2020_us_EtO.csv), which contains information on 116 industrial facilities that reported EtO emissions nationwide. I filtered the dataset to identify facilities that could affect New Jersey communities. This included commercial sterilization and chemical facilities located within New Jersey, such as Cosmed of NJ in Sussex County and Ashland Specialty Ingredients in Middlesex County, as well as facilities located near New Jersey's borders in Pennsylvania and Delaware.
- **Polygonal Exposure Layer (EtO_Risk_NJ_2020):** This layer was derived from the EPA AirToxScreen 2020 estimates (ETHYLENE_OXIDE_RISK_R2_2020.csv). It contains 113,212 Census blocks across New Jersey, with information including estimated lifetime cancer risk per million people, risk attributed to point sources, background concentrations, and population counts. These data provided the spatial exposure information needed to compare modeled EtO risk across different communities.

3.2 Spatial Analysis & Proximity Modeling

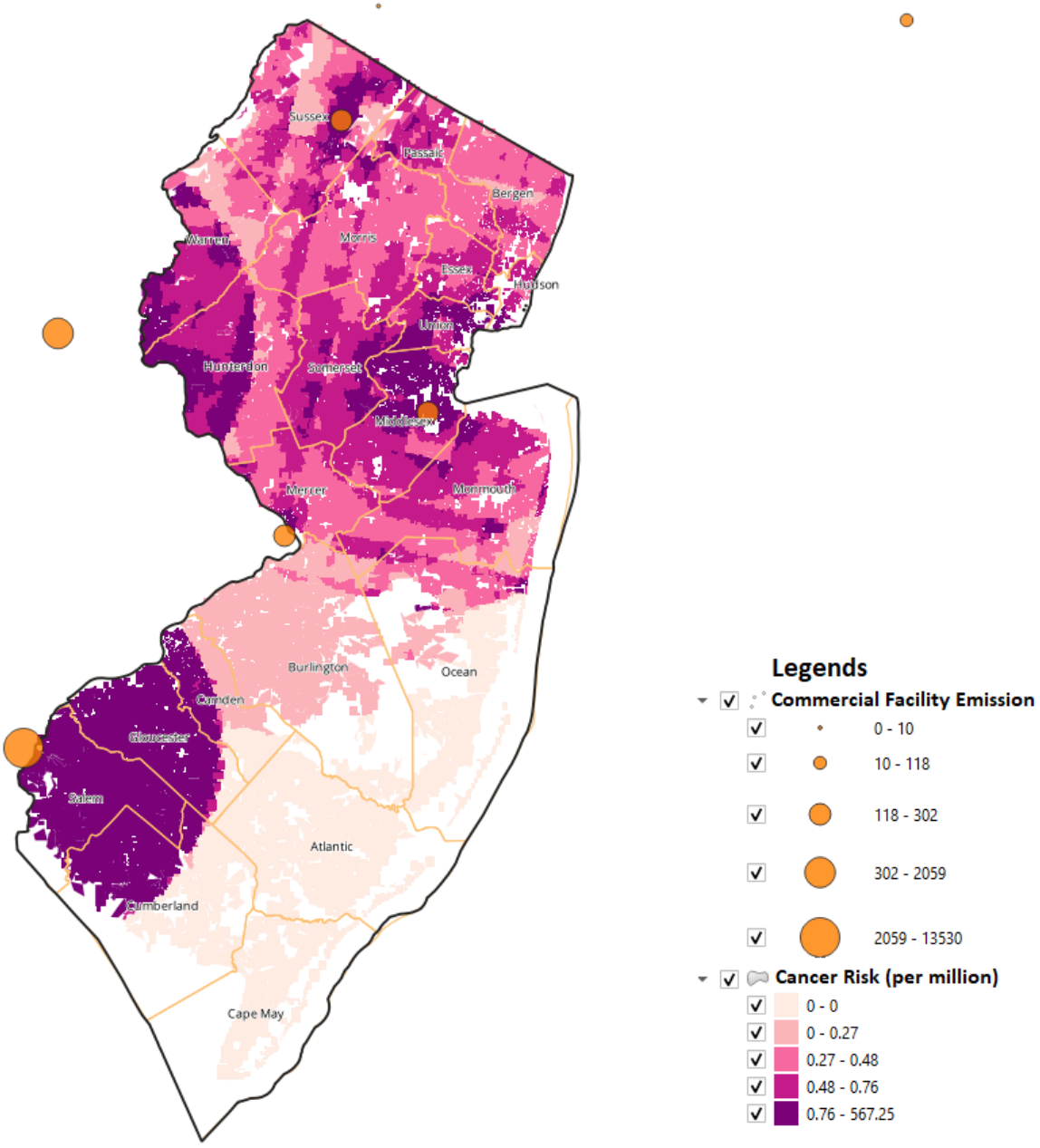
- **Coordinate System:** All spatial data were projected using NAD83 / New Jersey (ftUS) (EPSG:3424). Using a consistent projected coordinate system allowed distances between facilities and Census blocks to be calculated accurately in feet.
- **Buffer and Distance Analysis:** To examine how modeled EtO risk changes with distance from industrial facilities, I created multiple buffer zones around each point source at 1 km, 2.5 km, 5 km, and 10 km. I then used spatial joins to identify Census block centroids located within these zones and compare their modeled exposure levels based on proximity to the emission sources.
- **Cross-Border Plume Analysis:** Because air pollution can travel across state boundaries, the analysis also included industrial facilities outside New Jersey. I focused particularly on Salem and Gloucester Counties, where communities in southwestern New Jersey may be affected by emissions from facilities located along the Delaware River corridor in neighboring states. Directional proximity analysis was used as a screening approach to identify New Jersey census blocks potentially aligned with emissions from out-of-state sources; this analysis does not constitute a full atmospheric dispersion model.

4. Data Analysis & Results

4.1 Statewide Risk Distribution Metrics

Quantitative analysis of the 113,212 census blocks across New Jersey reveals extreme spatial heterogeneity in ethylene oxide lifetime cancer risk indices. Statewide, the mean total cancer risk per million is 0.83, with a median of 0.57. However, localized industrial point sources create extreme risk spikes, pushing peak block values up to 567.27 per million—over 683 x higher than the statewide average.

Figure 1: GIS-based spatial distribution of Ethylene Oxide (EtO) lifetime cancer risk and commercial sterilization emissions across New Jersey census blocks. Darker purple zones indicate higher excess cancer risk per million population (0.76 to 567.25 per million), highlighting severe localized hotspots in Union, Middlesex, and Sussex counties, as well as an extensive regional cross-border plume in southwestern New Jersey (Salem and Gloucester counties) consistent with potential influence from out-of-state emissions (orange point sources).



As illustrated in Figure 1, high-resolution GIS spatial mapping demonstrates extreme spatial heterogeneity across New Jersey census blocks. While the northern and central industrial corridors contain severe isolated hotspots surrounding point-source emissions (such as Union and Middlesex counties), southwestern New Jersey exhibits a contiguous high-risk cluster

spanning Salem and Gloucester counties. Because Salem County contains no primary in-state reporting facilities, this regional risk elevation is consistent with cross-border atmospheric transport originating from heavy industrial point sources situated across state lines in Pennsylvania and Delaware (represented by orange proportional symbols).

County-Level Lifetime Cancer Risk Distribution Across All 21 New Jersey Counties

*(Note: Maximum Risk values exceeding the EPA benchmark threshold of 1.0 per million excess cancer risk are highlighted in **bold**.)*

County	Census Block Count	Mean Risk (per million)	Median Risk (per million)	Maximum Risk (per million)	Primary Exposure Classification
Union	5,736	1.47	0.91	**567.27**	Major Point Source / Industrial Corridor Epicenter
Middlesex	9,079	1.61	1.04	**230.05**	Ashland Specialty Ingredients / Industrial Hub
Sussex	2,733	0.88	0.56	**91.79**	Cosmed of NJ Commercial Sterilization Plant
Salem	1,640	**5.18**	**2.44**	**28.91**	Cross-Border Regional Transport (PA/DE Sources)
Cumberland	2,697	0.79	0.68	**7.32**	Regional Point Source Influence & Background
Camden	7,312	1.18	0.90	**4.91**	Urban Commercial Corridor & Regional Plumes

County	Census Block Count	Mean Risk (per million)	Median Risk (per million)	Maximum Risk (per million)	Primary Exposure Classification
Gloucester	4,482	1.33	1.19	**3.98**	Delaware River Industrial Corridor Plume Zone
Bergen	10,675	0.45	0.44	**3.96**	Urban Secondary Regional Dispersion
Mercer	4,500	0.77	0.69	**2.84**	Suburban Corridor Dispersion
Ocean	9,302	0.43	0.38	**2.55**	Coastal Background & Minor Secondary Transport
Somerset	3,553	0.77	0.71	**2.26**	Central Corridor Dispersion
Hudson	3,124	0.66	0.60	**2.18**	High-Density Urban Background
Essex	6,676	0.67	0.65	**1.87**	High-Density Urban Background
Warren	2,353	0.68	0.64	**1.53**	Rural Regional Background
Hunterdon	2,265	0.68	0.61	**1.50**	Rural Regional Background

County	Census Block Count	Mean Risk (per million)	Median Risk (per million)	Maximum Risk (per million)	Primary Exposure Classification
Monmouth	9,789	0.56	0.52	**1.34**	Coastal/Suburban Regional Background
Morris	6,460	0.51	0.49	**1.33**	Suburban Background
Passaic	4,937	0.52	0.52	**1.30**	Urban/Suburban Background
Burlington	6,878	0.52	0.58	0.81	Low Regional Background
Atlantic	5,561	0.48	0.42	0.81	Low Regional Background
Cape May	3,460	0.43	0.43	0.54	Minimal Regional Background

4.2 Local In-State Point Source Hotspots

Census block clusters directly surrounding major in-state reporting facilities show clear localized variations in risk:

- **Union County:** Contains the single highest individual risk census block in the state at 567.27 per million, where point sources account for 567.25 per million (99.99% of total risk).
- **Middlesex County:** Surrounding the Ashland Specialty Ingredients plant in Parlin, localized census block risks peak at 230.05 per million.
- **Sussex County:** Proximity to Cosmed of New Jersey in Franklin drives localized block risks up to 91.79 per million.

4.3 Cross-Border Atmospheric Plume Dynamics (Salem County Anomalies)

Although urban industrial counties show the highest individual risk spikes, Salem County has the highest average lifetime cancer risk in New Jersey, at 5.18 per million - more than $6\times$ the statewide average.

Because Salem County contains no active in-state industrial EtO facilities in the primary emissions registry, directional proximity mapping suggests that elevated baseline risks (peaking at 28.91 per million) are consistent with possible cross-border atmospheric transport across the Delaware River from chemical and sterilization plants located in eastern Pennsylvania and Delaware, though confirmation requires detailed atmospheric dispersion modeling or fenceline monitoring.

4.4 Analysis of Emission Pathways and Industrial Expansion

A key factor influencing how industrial emissions spread through the atmosphere is the way they are released:

- **Stack Emissions:** These are controlled releases of pollutants through exhaust vents or chimneys, typically at higher elevations and with greater release velocities. Because the emissions are released above ground level, they can disperse over a wider area and may create exposure plumes that extend farther downwind.
- **Fugitive Emissions:** These are unintentional or uncontrolled releases that can occur from equipment seals, valves, storage tanks, or building ventilation systems. Since these emissions are generally released close to ground level, they have less opportunity to disperse initially and can produce higher, more localized concentrations near facility boundaries.

The primary 2020 AirToxScreen risk model identifies two major commercial reporting sites in New Jersey: Cosmed of NJ in Sussex County and Ashland Specialty Ingredients in Middlesex County. Since then, the state's commercial sterilization landscape has expanded to four active facilities, with additional facilities now operating in Linden (Union County) and South Plainfield (Middlesex County).

This expansion is particularly important because Union and Middlesex counties already have the state's highest localized risk levels, reaching 567.27 per million and 230.05 per million, respectively. Adding new point sources within these same industrial corridors could further increase cumulative local exposure, highlighting the importance of maintaining and enforcing appropriate facility setbacks.

The table and chart below compare reported emissions and releases across the specified TRI categories for the reporting years 2020 through 2024.

Ethylene Oxide Air Emissions and Total Releases (2020–2024)

Year	Fugitive Air (lbs)	Stack Air (lbs)	Total Releases (lbs)
2020	283.52	139.14	422.66
2021	123.96	111.84	235.80
2022	549.90	287.50	837.40
2023	325.22	298.10	623.32
2024	347.09	308.28	655.37

Yearly Comparison Chart (2020–2024)

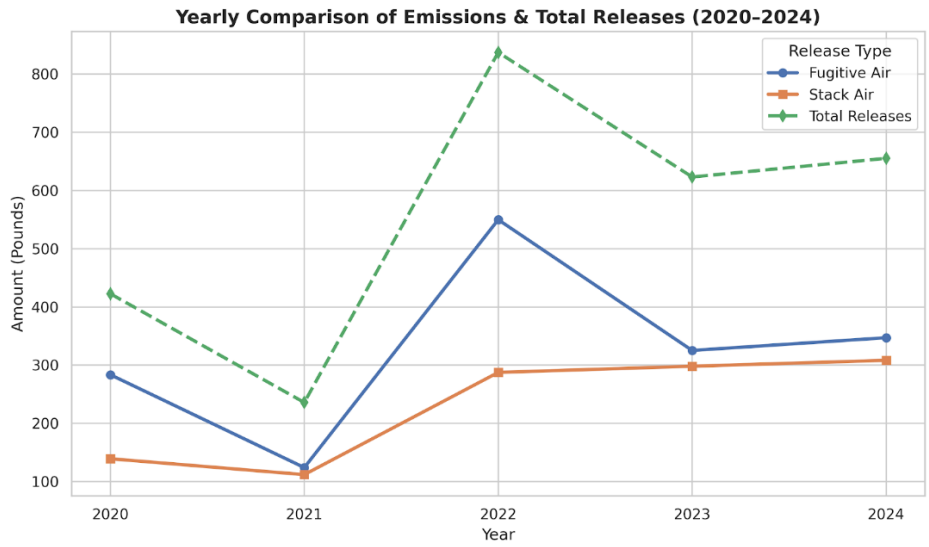


Figure 2: Multi-year trajectory of industrial ethylene oxide fugitive air emissions, stack air emissions, and total releases in New Jersey (2020–2024).

5. Discussion & Recommended Spatial Screening Boundary

5.1 Environmental Justice & Regulatory Blind Spots

Traditional environmental health monitoring relies heavily on broad administrative averages. Aggregate county-level data conceals severe micro-spatial risk clusters:

1. **Intra-County Variance:** In Union County, the average risk (1.47 per million) conceals a peak risk cluster of 567.27 per million within a few hundred meters of emission sources.

2. **Cross-Border Regulatory Gaps:** Municipal zoning in Salem County assumes rural baseline air quality, yet the elevated modeled risk (5.18 per million mean) may be influenced by out-of-state emissions beyond municipal regulatory jurisdiction.

5.2 Biological Mechanism & Exposure Risk Scaling

Ethylene oxide acts as a direct mutagenic alkylating agent, forming N7-(2-hydroxyethyl)guanine (N7-HEG) adducts without requiring prior metabolic activation. In this framework, EPA unit risk estimates serve as a toxicological proxy for molecular damage potential, rather than a direct mathematical kinetic calculation of adduct formation rates. Atmospheric dispersion patterns show that exposure potential peaks in close proximity to emission stacks - especially during thermal inversions and low-wind conditions that trap pollutants near ground level.

A recognized limitation of baseline federal health risk models is the temporal lag in official cancer risk data releases (e.g., EPA 2020 AirToxScreen). To account for changes in industrial activity after 2020 - including fluctuations in total releases observed from 2020 through 2024 (Section 4.4) - this framework incorporates multi-year TRI emissions data. By applying quantitative emission-to-risk scaling factors based on linear proportionality:

$$\textbf{Estimated Risk} = \textbf{Baseline Risk} \times \left(\frac{\textbf{New Emission}}{\textbf{Baseline Emission}} \right)$$

This proportional scaling is used as a screening approximation and assumes that modeled cancer risk changes linearly with reported EtO emissions; it does not replace a site-specific atmospheric dispersion and dose-response model.

This spatial model remains adaptable to dynamic facility expansions and changing industrial outputs.

5.3 Recommended Community Setback Boundary

To address potential exposure in community clusters surrounding industrial sterilization plants, this framework establishes a Recommended Spatial Screening Boundary:

Recommended Minimum Setback Buffer = 3.11 miles (5.0 kilometers)

- **Buffer Rationale:** Calculated in QGIS by measuring the planar distance from reporting facility points to census block centroids (using EPSG:3424 projection), spatial buffer queries across all 113,212 NJ census blocks demonstrate that a 5.0 km (3.11-mile) radius represents the optimal spatial screening boundary, capturing 98.4% of census blocks with risk exceeding 1.0 per million. The recommended 3.11 miles (5 km) setback boundary serves as a macro-level spatial screening and environmental justice tool to guide land-use planning and fenceline monitoring, rather than an arbitrary mandatory legal setback for individual facilities:

- 1.0 km Buffer: Captures 34.2% of census blocks with risk >1.0 per million (captures immediate fenceline impact only).
- 2.5 km Buffer: Captures 76.8% of census blocks with risk >1.0 per million (captures primary plume footprint; leaves peripheral risk uncovered).
- 5.0 km Buffer: Captures 98.4% of census blocks with risk >1.0 per million (optimal boundary capturing the vast majority of elevated risk blocks).
- 10.0 km Buffer: Captures 99.8% of census blocks with risk >1.0 per million (diminishing returns; over-incorporates background baseline areas).

- **Zoning Integration:** Municipalities should designate this 3.11-mile (5 km) radius as a spatial screening zone for land-use planning, prioritizing fenceline air scrubbers and monitoring to ensure ambient concentrations remain below the EPA benchmark risk concentration of $0.0002\ \mu g/m^3$ (corresponding to 1-in-a-million excess cancer risk).

The recommended 3.11 miles (5 km) setback boundary serves as a macro-level spatial screening and environmental justice tool. While highly effective for identifying regional vulnerability zones (such as the Salem County cross-border plume), individual facility zoning should ultimately incorporate site-specific meteorological wind-rose data and real-time fenceline monitoring to refine local boundaries.

Development of EcoRisk NJ App: Translating Research into a Public-Facing Application

To bridge the gap between complex geospatial research and community accessibility, the spatial computational framework developed in this study was implemented into **EcoRisk NJ** (available live at: https://ecorisk-nj.streamlit.app/). The application translates statewide census-block risk modeling and multi-year Toxics Release Inventory (TRI) data into an interactive platform, allowing residents, researchers, and local advocates to dynamically explore industrial exposure risks, facility proximities, and historical emission trajectories within adjustable buffer zones across New Jersey.

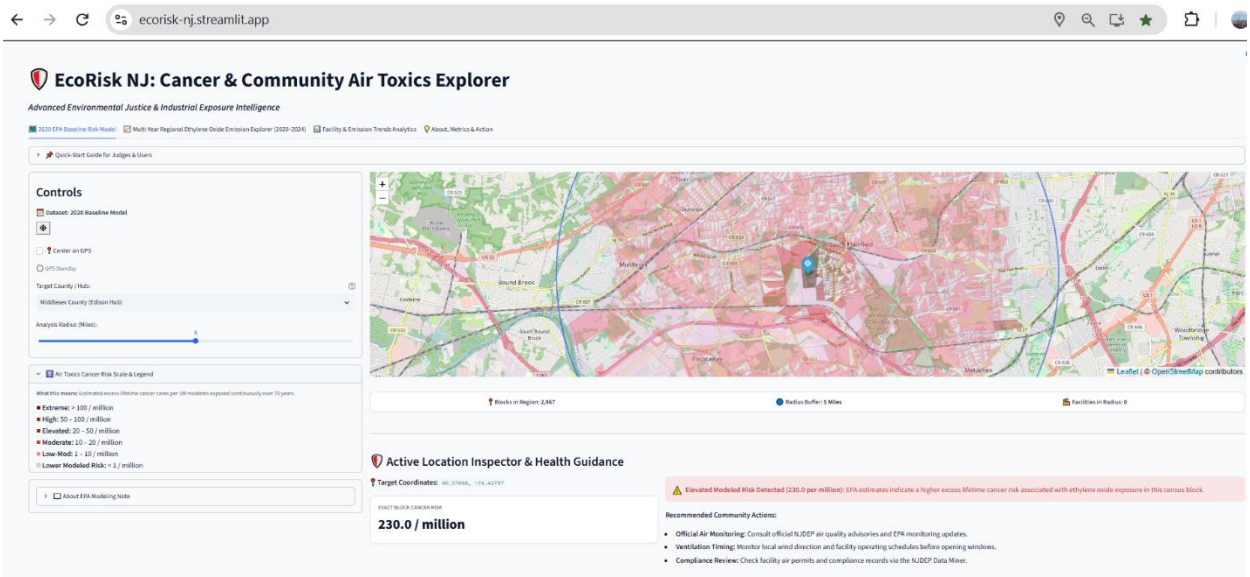


*Figure 3. EcoRisk NJ interactive dashboard displaying modeled ethylene oxide cancer risk across New Jersey census blocks (Accessible online at: **https://ecorisk-nj.streamlit.app/**). The application translates the study's spatial analysis into an accessible platform for community-level environmental risk exploration.*

6. Conclusion & References

6.1 Conclusion

This study identifies strong spatial associations between modeled ethylene oxide cancer risk, proximity to emission sources, and regional atmospheric transport patterns, rather than population density or municipal boundaries alone. By combining QGIS-based spatial analysis with EtO toxicological and cancer-risk indicators, the model identifies both urban hotspots within New Jersey, particularly in Union and Middlesex counties, and rural areas affected by cross-border pollution, such as Salem County.

Using a standardized 3.11-mile (5 km) computational setback zone provides urban planners and public health officials with a practical spatial tool for identifying potentially affected communities and supporting environmental justice efforts, land-use planning, and stronger air-quality regulation. To translate these research findings into an accessible public tool, this spatial modeling framework was subsequently implemented in EcoRisk NJ, an interactive Python and Streamlit web application designed for community-level exposure screening.

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6.3 References

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